HEALTH360

Smart Healthcare Analytics

Patient Risk & Lifestyle Insights

PROJECT REPORT

A visual analytics project presenting healthcare indicators and lifestyle insights through a Flask-based website and an interactive Tableau dashboard.

Python • Flask • HTML • CSS • JavaScript • Tableau Public

2026

PROJECT LINKS

The following links were supplied with the project materials:

GitHub Repository:
https://github.com/siddheshmk4836-prog/Health-360-flask

Tableau Public Dashboard / Story:
https://public.tableau.com/views/Health360_ComprehensiveHealthcareRiskDiseaseAnalytics/SmartHealthcareAnalyticsPatientRiskLifestyleInsights?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Video Presentation:
https://drive.google.com/drive/folders/1jb3WbLd-8LY7EpsnL5pPnCLve28849iQ?usp=sharing

Dataset link: Not included in the supplied material. Add the original dataset URL here before final submission, if required.

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1. INTRODUCTION

1.1 Project Overview

Health360 is a healthcare analytics project that presents patient and lifestyle information through an interactive dashboard and story. The supplied Tableau visuals organize the analysis into key performance indicators and charts covering patient counts, diabetes, heart disease, stroke, exercise level, blood pressure, alcohol and smoking consumption, BMI categories, age groups, and fatigue condition.

The project combines a Python Flask website with web technologies for the application interface and Tableau Public for interactive data visualization. The GitHub repository contains the submitted application source and supporting project files.

1.2 Purpose

• To make summarized healthcare and lifestyle information easier to explore through visual analytics.

• To compare health indicators across age groups, BMI categories, and exercise-level categories.

• To provide a consolidated view of selected risk-related indicators for learning, analysis, and demonstration.

• To present the project through a web application, Tableau dashboard, source-code repository, and video presentation.

Scope note: The dashboard is an analytical and educational visualization. It is not a clinical diagnostic tool and should not be used to make individual medical decisions.

2. IDEATION PHASE

2.1 Problem Statement

Healthcare datasets can contain multiple variables that are difficult to interpret when viewed only as rows and columns. Users may find it challenging to compare patient indicators across age groups and lifestyle categories. Health360 addresses this presentation problem by arranging selected measures into a dashboard with charts, labels, summary KPIs, and a navigable story.

2.2 Empathy Map Canvas

Think & Feel	Wants a quick, understandable overview of patient and lifestyle patterns.
See	Sees multiple health measures and categories in a dataset or dashboard.
Say & Do	Selects categories, compares charts, and explores age/BMI/lifestyle patterns.
Pains	Raw tables can be lengthy; comparisons across multiple variables may take time.
Gains	A consolidated visual view, clearer comparisons, and accessible project presentation.

This empathy map is a project-design interpretation of the intended dashboard user; it is not a record of interviews or a formal user study.

2.3 Brainstorming

• Use KPI cards to show the overall patient count and selected condition counts.

• Use a donut chart for active versus sedentary exercise-level distribution.

• Compare average blood pressure across age groups.

- Compare heart-disease rates across BMI categories and exercise levels.
- Compare alcohol and smoking measures across age groups.
- Use a story/dashboard navigation experience to make the visual analysis easier to present.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User activity	System / dashboard response
Access	Opens the Health360 website or Tableau view.	Displays the project landing/dashboard experience.
Orient	Reviews the KPI overview.	Shows total patients and selected condition counts.
Explore	Chooses a chart or story section.	Displays the corresponding visual analysis.
Compare	Compares age, BMI, exercise, and lifestyle categories.	Uses grouped bars, pie/donut charts, and heatmap-style table.
Review	Checks patterns and summary findings.	User interprets displayed values and limitations.

3.2 Solution Requirement

• A Python Flask application to serve the website pages.

• HTML templates for the base layout, dashboard, story, and Tableau view.

• CSS and JavaScript assets for page styling and front-end behavior.

• A dataset containing the patient and lifestyle fields used in the visualizations.

• Tableau worksheets/dashboard/story for the chart-based analysis.

• A modern web browser and internet connection to access the hosted Tableau Public visualization.

3.3 Data Flow Diagram

Input Patient / lifestyle dataset	→	**Processing** Data preparation, grouping, aggregation, KPI calculations	→	**Output** Charts, KPIs, dashboard and story
User Opens website / selects view	→	**Flask application** Routes and HTML templates	→	**Visualization** Tableau Public view

Conceptual data flow based on the submitted project description and visible outputs. Exact dataset ingestion and backend processing details should be aligned with the implementation source code.

3.4 Technology Stack

Technology	Role in project
Python	Backend programming language.
Flask	Web framework used to serve application routes and pages.
HTML	Structure of website pages and templates.
CSS	Website styling and layout.
JavaScript	Client-side interactions and dashboard-related front-end behavior.
Tableau Public	Interactive charts, dashboard, and story visualization.
Git / GitHub	Version control and source-code hosting.

4. PROJECT DESIGN

4.1 Problem-Solution Fit

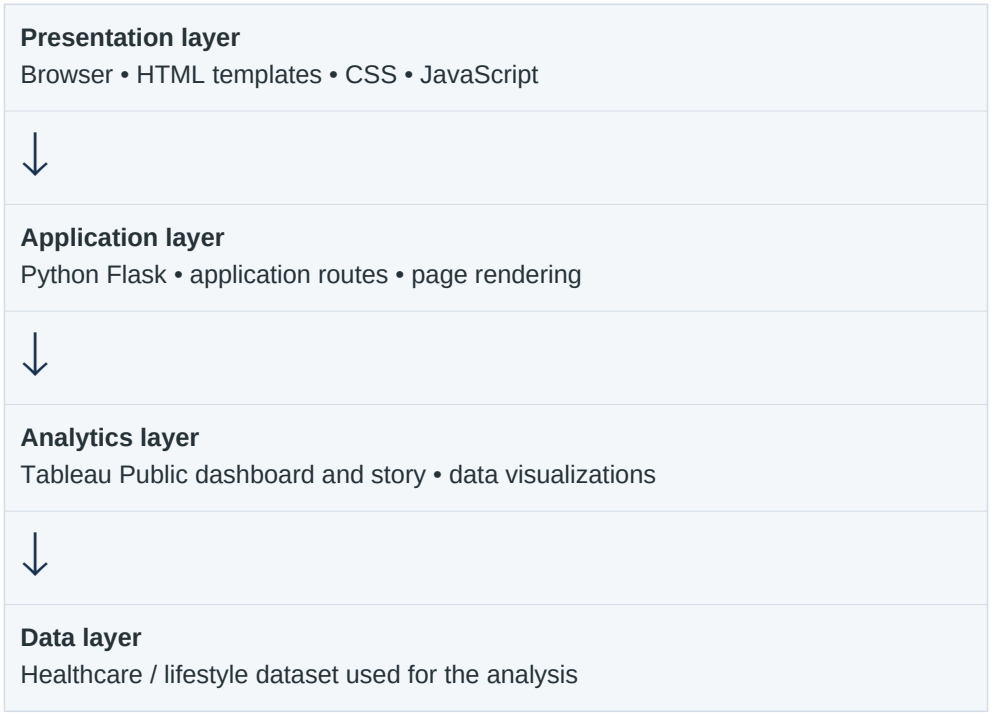
The project focuses on the gap between raw health-related records and an easy-to-review visual summary. KPI cards support a quick overview, while charts allow category-level comparisons. The dashboard and story format are suitable for users who need to review several related indicators in one place.

4.2 Proposed Solution

Health360 provides a Flask-based website structure and a Tableau Public visualization experience. The Tableau view presents a KPI overview and multiple analytical sheets. The screenshots supplied for this report show the following visual components:

- KPI Overview: Total Patients, Diabetes, Heart Disease, and Stroke.
- Exercise Level Distribution: Active and Sedentary categories.
- Average Blood Pressure Across Age Groups.
- BMI Category vs Heart Disease Rate.
- Heart Disease by BMI Category and Exercise Level.
- Heart Disease vs Diabetes Prevalence by BMI Category.
- Alcohol and Smoking Consumption Across Age Groups.
- Patient Distribution by Age Group.
- Age Group Distribution by Fatigue Condition.

4.3 Solution Architecture



Architecture is a high-level representation of the supplied project. The exact method used to connect Flask pages to Tableau (for example, embedding or linking) should be confirmed against the running application.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activities	Deliverable
1. Ideation	Identify the need for a consolidated healthcare analytics view; define dashboard topics.	Problem definition and feature list.
2. Requirements	Identify the dataset fields, application pages, visualization needs, and tools.	Requirement outline and technology stack.
3. Design	Plan page structure, dashboard sections, chart types, and navigation.	Website and dashboard design.
4. Development	Build Flask application structure, templates, CSS/JS assets, and Tableau visuals.	Working application and Tableau view.
5. Testing	Open pages, inspect chart rendering, verify links, and review displayed values.	Tested project build.
6. Deployment / submission	Push source code to GitHub and prepare video presentation and report.	Repository, demo video, and report.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The supplied evidence consists of Tableau screenshots and the GitHub repository link. The checks below describe practical functional and visual checks for the submitted build; no measured load-time, throughput, or stress-test results were supplied.

Test area	Check performed / recommended	Expected result
Flask pages	Open the home/dashboard/story/Tableau routes in a browser.	Each configured route loads the intended page.
Static assets	Check CSS and JavaScript loading in browser developer tools.	No missing static files; layout and interactions render.
Tableau view	Open the public dashboard/story link.	Visualization loads and charts are readable.
Chart values	Compare displayed KPI and chart values with the source dataset / Tableau workbook.	Displayed aggregates match the workbook calculations.
Navigation	Use the website links and dashboard/story navigation.	Links open the correct destination.
Responsive layout	Test at desktop and mobile viewport widths.	Content remains usable without clipping.
Repository	Open the GitHub repository and inspect committed files.	Project source and supporting files are accessible.

Performance metrics: No formal response-time, concurrent-user, CPU, memory, or stress-test measurements were provided. These should be measured and recorded if the evaluation rubric requires quantitative performance testing.

7. RESULTS

7.1 Output Screenshots

The screenshots below are taken from the Tableau workbook/dashboard materials supplied for this report. The values are reproduced as visible in those screenshots.

KPI overview: The dashboard shows 5,000 total patients, 1,746 diabetes cases, 1,499 heart-disease cases, and 1,243 stroke cases. These are displayed dataset counts, not population prevalence estimates.

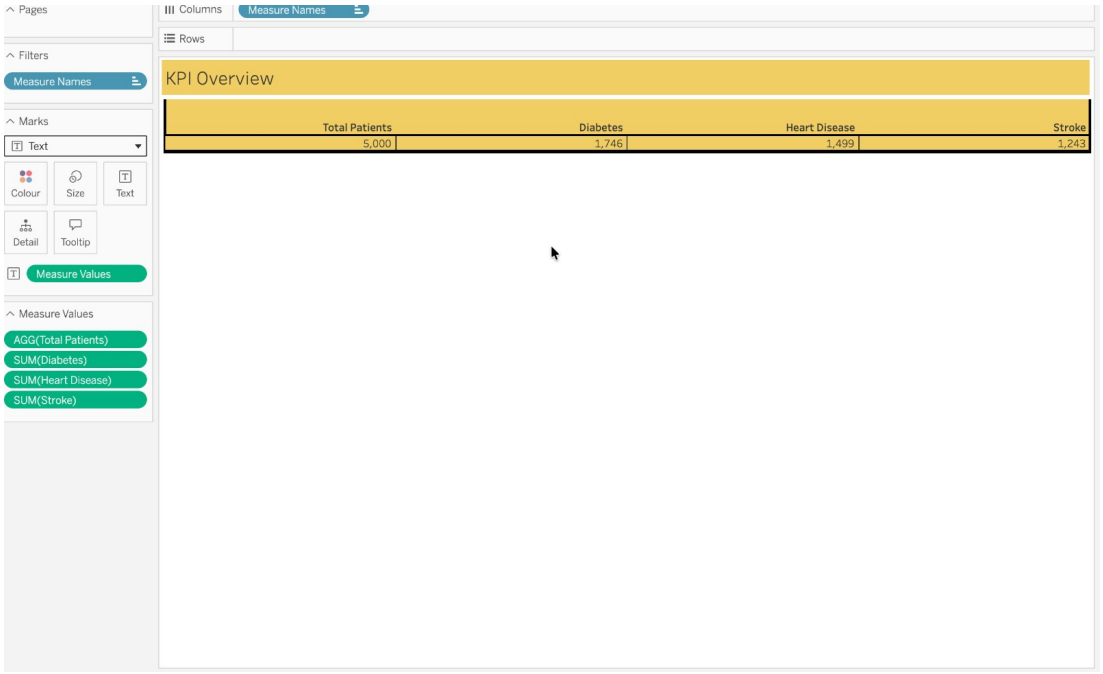


Figure 1. KPI Overview

Exercise level distribution: The donut chart shows 67.06% Active and 32.94% Sedentary.

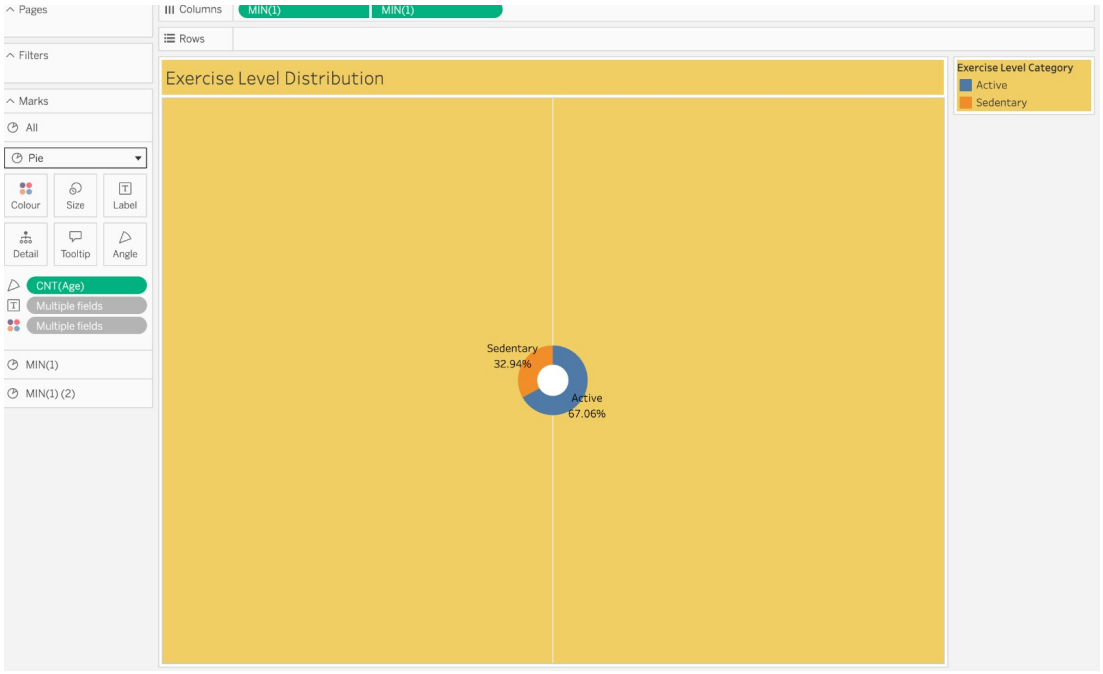


Figure 2. Exercise Level Distribution

7. RESULTS — VISUAL ANALYSIS

Average blood pressure across age groups: The visible averages range from approximately 133.610 to 135.665. The Old group has the highest displayed average (135.665), while Adult has the lowest (133.610).



Figure 3. Average Blood Pressure Across Age Groups

BMI category vs heart disease rate: The chart displays Obese 39.98%, Overweight 32.91%, Normal 25.97%, and Underweight 20.54%.



Figure 4. BMI Category vs Heart Disease Rate

7. RESULTS — VISUAL ANALYSIS (CONTINUED)

Alcohol and smoking across age groups: In the supplied chart, the Teenager category has the largest displayed values for both Alcohol (514) and Smoking (526). The chart uses aggregated values; the precise unit/definition of these measures should be checked in the dataset/workbook.



Figure 5. Alcohol and Smoking Consumption Across Age Groups

Heart disease by BMI and exercise level: The heatmap displays the highest shown average heart-disease value for Obese/Sedentary (0.4286), followed by Obese/Active (0.3881).

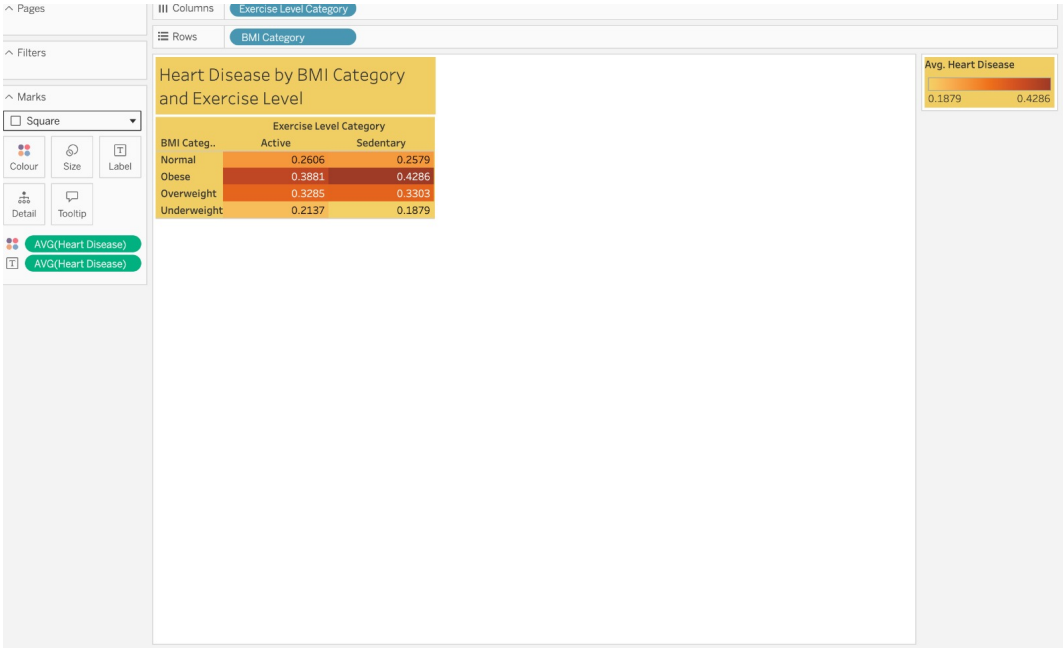


Figure 6. Heart Disease by BMI Category and Exercise Level

7. RESULTS — VISUAL ANALYSIS (CONTINUED)

Heart disease vs diabetes by BMI category: The grouped bars compare the two measures across Normal, Obese, Overweight, and Underweight BMI categories.



Figure 7. Heart Disease vs Diabetes Prevalence by BMI Category

Patient distribution by age group: The pie chart shows Teenager 20.38%, Adult 16.26%, Senior 16.38%, Middle Age 16.08%, Young Adult 15.86%, and Old 15.04%.

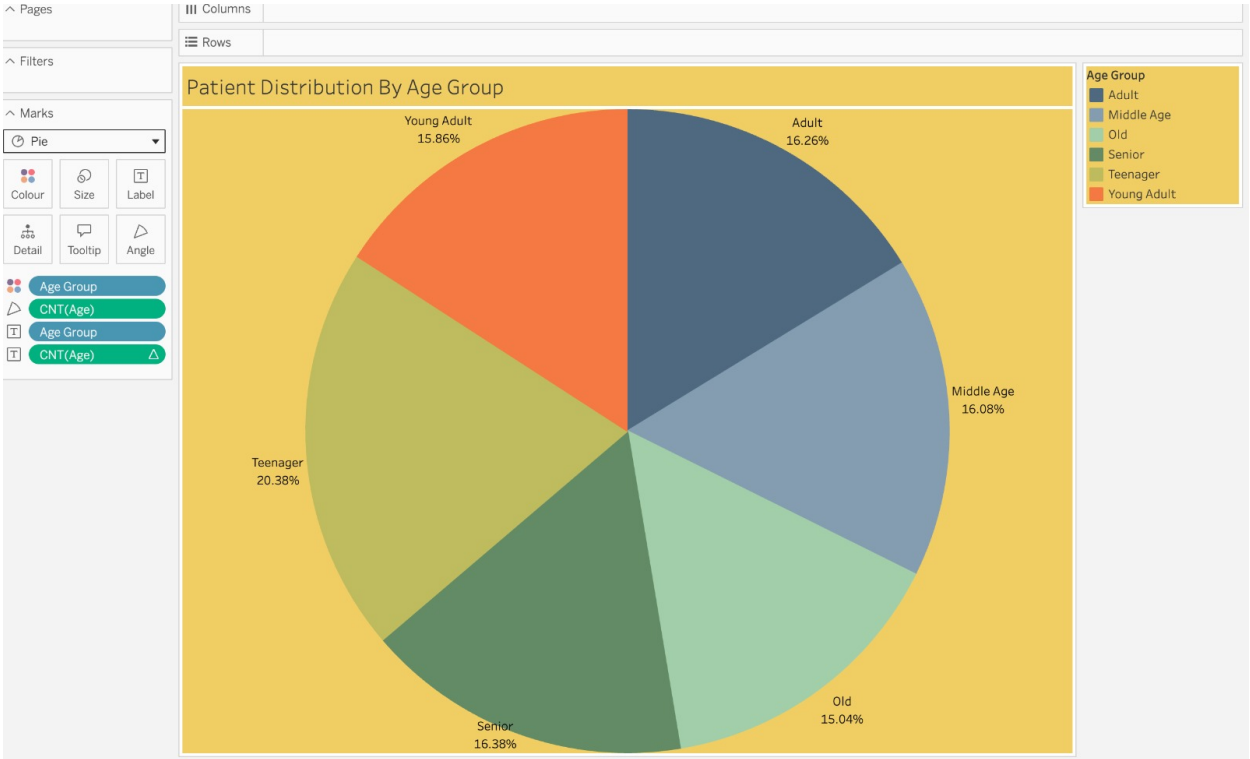


Figure 8. Patient Distribution by Age Group

7. RESULTS — DASHBOARD AND STORY

Age group distribution by fatigue condition: The stacked horizontal bar chart separates the two fatigue-condition categories for each age group. The labels in the supplied screenshot show the two groups as 1 and 0; their exact coding meaning should be confirmed from the dataset documentation.



Figure 9. Age Group Distribution by Fatigue Condition

Dashboard composition: The overview screenshot combines the KPI strip with multiple charts in a single dashboard canvas.

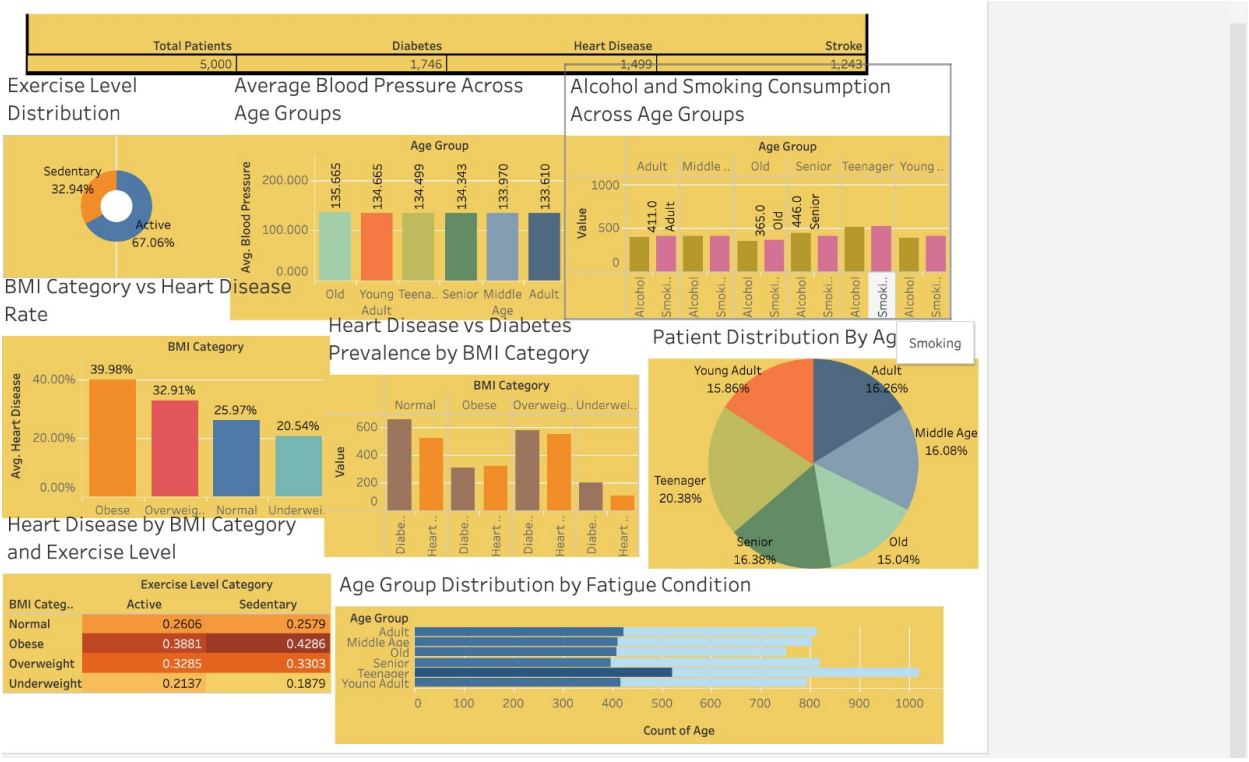


Figure 10. Smart Healthcare Analytics Dashboard Overview

Story presentation: The story screenshot shows navigation between analytical views and presents the alcohol/smoking chart as the selected story point.

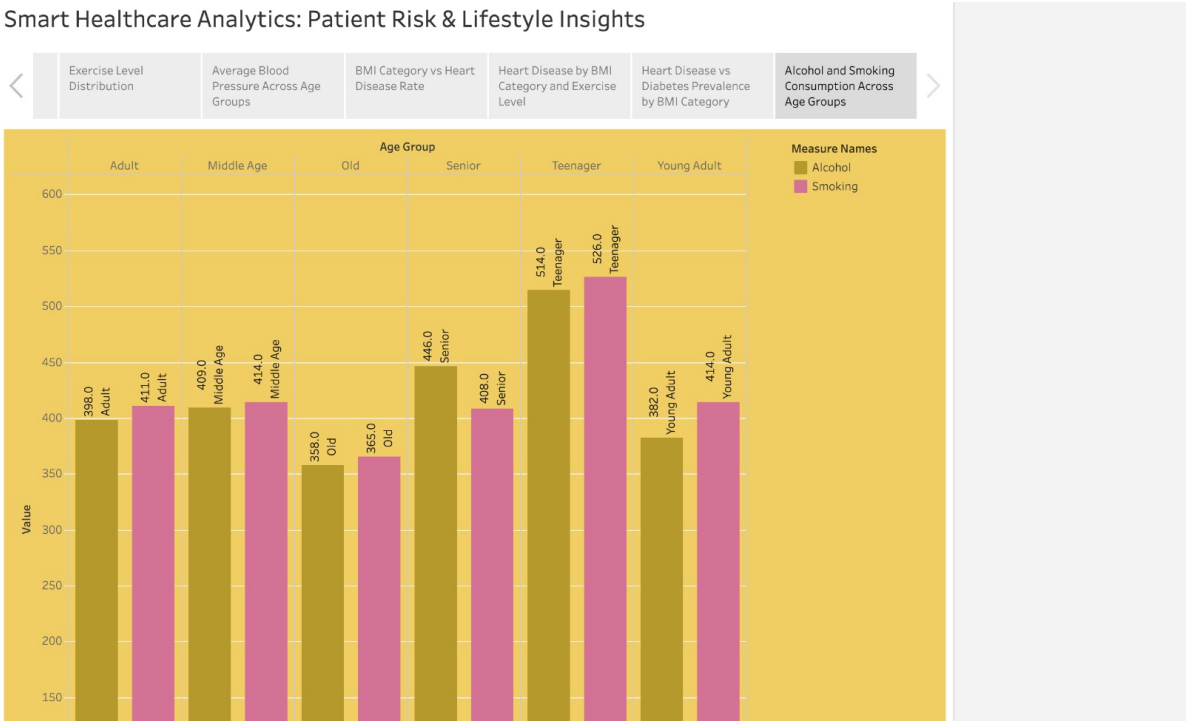


Figure 11. Tableau Story View

8. ADVANTAGES & DISADVANTAGES

Advantages

• Combines several health and lifestyle indicators in one visual analytics experience.

• KPI summary helps users understand the overall dataset at a glance.

• Charts support comparisons across age groups, BMI categories, and exercise levels.

• Tableau Public makes the visualization accessible through a shareable web link.

• Flask, HTML, CSS, and JavaScript provide a customizable website layer.

• GitHub provides version history and a central location for project source code.

Disadvantages / Limitations

• The accuracy and usefulness of findings depend on the quality, representativeness, and documentation of the underlying dataset.

• The screenshots do not provide enough information to verify the dataset source, collection method, or definitions of every field.

• Some visual measures (for example, alcohol/smoking totals and fatigue codes) need clearer units or category definitions.

• A public Tableau visualization generally requires an internet connection to load.

• No formal performance benchmark or user evaluation results were supplied.

• The dashboard shows associations and summaries; it does not establish causation or provide medical diagnoses.

9. CONCLUSION

Health360 demonstrates how a Flask-based web application and Tableau visualizations can be used to present healthcare and lifestyle data in an accessible format. The submitted dashboard summarizes 5,000 patient records and provides views of diabetes, heart disease, stroke, exercise level, blood pressure, BMI, age distribution, fatigue, alcohol, and smoking. The project deliverables include the GitHub source repository, Tableau Public visualization, and video presentation link. The work provides a foundation for further improvements in data documentation, usability, testing, and deployment.

10. FUTURE SCOPE

• Add clear dataset provenance, field definitions, units, and data-cleaning documentation.

• Add interactive filters for age group, BMI category, exercise level, and other relevant dimensions.

• Improve responsive design and accessibility of the Flask website.

• Add user authentication and role-based access if private or sensitive data is introduced.

• Deploy the Flask application to a suitable hosting platform and document the deployment process.

• Add automated data validation and reproducible data-processing steps.

• Conduct load testing and usability testing, and record quantitative results.

• Add responsible-use guidance and privacy safeguards before using real patient information.

11. APPENDIX

A. Source Code

The project source code is available in the GitHub repository:

https://github.com/siddheshmk4836-prog/Health-360-flask

B. Dataset Link

The dataset URL was not present in the supplied materials. Insert the original dataset link here before submission. Do not substitute the Tableau visualization link for the dataset source.

C. GitHub & Project Demo Link

GitHub: Health360 Flask Repository

Tableau Public Dashboard / Story: Open Tableau Public Visualization

Video Presentation: Open Video Presentation Folder

End of Report